Reference Notes

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Part I

Readings & Literature Review

1. Age-Related Macular Degeneration: A Review

Overview

Age-related macular degeneration (AMD) is a progressive degenerative disease of the macula, the central retinal region responsible for high-acuity vision. AMD is among the leading causes of irreversible vision loss in adults over age 55 and is projected to affect nearly 288 million individuals worldwide by 2040.

The disease primarily affects photoreceptors, retinal pigment epithelium (RPE), bruch membrane, and choriocapillaris. AMD progression involves degeneration of these structures, ultimately leading to central vision loss. Early and intermediate disease may be asymptomatic, while advanced disease can substantially impair reading, driving, facial recognition, and other daily activities.

Pathophysiology

AMD is considered a multifactorial disease driven by interactions among aging, genetic susceptibility, environmental exposures, inflammation, and metabolic dysfunction.

Major biological processes implicated include chronic inflammation, complement pathway dysregulation, oxidative stress, lipid and lipoprotein accumulation, impaired extracellular matrix maintenance and choriocapillaris degeneration. Normal aging causes reduced photoreceptor density lipid deposition within Bruch membrane reduced density of choriocapillaris vessels and increased vascular resistance

These alterations disrupt retinal homeostasis and contribute to extracellular deposit formation. The hallmark lesion of AMD is the druse (plural: drusen), consisting of lipids, proteins, and minerals accumulating between the RPE and Bruch membrane.

Classification of AMD

Stage	Defined By	Clinical Characteristics	OCT Findings	Notes
Early AMD	Medium drusen ($> 63\,\mu m$ and $\leq 125\,\mu m$) No AMD-associated pigmentary abnormalities	Usually asymptomatic Visual acuity often preserved	Small dome-shaped RPE elevations corresponding to drusen Relatively preserved retinal architecture	Progression risk is relatively low at this stage.
Intermediate AMD	Large drusen ($> 125\,\mu m$) And/or pigmentary abnormalities	Difficulty in low-light conditions Reduced contrast sensitivity Metamorphopsia Mild visual blurring	Larger drusen elevations Hyperreflective foci Pigment migration Structural irregularities of the RPE	Most important stage for prediction studies because progression to advanced disease frequently occurs from this state.
Late AMD: Geographic Atrophy	Confluent regions of photoreceptor loss RPE loss Choriocapillaris degeneration	Progressive central vision loss Central scotomas Declining visual acuity	Loss of outer retinal layers RPE attenuation or absence Increased signal transmission into the choroid Hypertransmission defects	Geographic atrophy enlarges over time and is associated with irreversible visual loss.
Late AMD: Neovascular	Growth of abnormal blood vessels beneath or within the retina VEGF-mediated angiogenesis	Often rapid symptom onset Sudden vision decline Metamorphopsia Central visual defects	Intraretinal fluid Subretinal fluid Pigment epithelial detachments Hemorrhage-associated abnormalities	Responsible for most severe acute vision loss associated with AMD.

Table 1: Clinical classification of age-related macular degeneration (AMD).

Major Risk Factors

Age

Age is the strongest risk factor.

The incidence of late AMD increases dramatically with age:

- 0.3 per 1000 individuals aged 55–59
- 5.7 per 1000 individuals aged 75–79
- 36.7 per 1000 individuals aged 90+

Genetics

Estimated heritability is approximately 71%.

Major genetic loci include CFH and ARMS2-HTRA1, which are involved in complement regulation, inflammation, lipid metabolism, and extracellular matrix maintenance.

Smoking

Smoking is the strongest established environmental risk factor. Studies consistently demonstrate substantially increased AMD incidence among heavy smokers.

Diet and Lifestyle

Protective factors include fish consumption, fruits and veggies, and physical activity.

Features Relevant to Progression Prediction

For a progression-prediction project, the review repeatedly highlights several structural features associated with worsening disease.

Drusen Burden

Important characteristics include the number of drusen, total area, volume, and presence of large drusen. Increasing drusen burden characterizes progression from early to intermediate AMD.

Pigmentary Abnormalities

Pigment changes arise from migration of RPE cells into inner retinal layers.

These appear as:

- Hyperpigmentation on fundus imaging
- Hyperreflective foci on OCT

The review identifies pigmentary abnormalities as a major marker of progression risk.

Hyperreflective Foci

Migration of RPE cells creates hyperreflective structures visible on OCT.

These likely reflect active retinal remodeling and degeneration.

Outer Retinal Degeneration

Progressive loss of photoreceptors, ellipsoid zone integrity, and RPE structure may preced advanced disease development.

Hypertransmission Defects

Loss or attenuation of the RPE permits increased OCT signal penetration into deeper tissue.

These appear as vertical transmission columns extending into the choroid.

Such findings are particularly relevant to the current project because the proposed "barcoding" phenomenon appears visually related to localized transmission abnormalities beneath regions of RPE degeneration.

Geographic Atrophy Precursors

Potential indicators of future atrophy include RPE disruption, hypertransmission, outer retinal thinning, or progressive drusen remodeling.

High-Risk Clinical Profile

The AREDS study reported that:

- 25.9% of individuals with large drusen and pigment abnormalities progressed to late AMD within 5 years
- If one eye already had advanced AMD, progression in the fellow eye reached 53.1% within 5 years when large drusen and pigment abnormalities were present

These findings make large drusen and pigmentary abnormalities particularly important baseline predictors.

2. The Barcode Sign in Optical Coherence Tomography

Overview

El Zawahry et al. described a characteristic OCT imaging pattern termed the *barcode sign*, observed in eyes with extensive corneal neovascularization. The barcode sign appears as multiple vertically oriented dark stripes extending through the OCT image. These stripes arise from attenuation of the OCT signal caused by structures that strongly block light transmission. Although the study focuses on corneal vascularization rather than age-related macular degeneration (AMD), it provides an important example of how biological structures can generate stripe-like OCT patterns through optical shadowing mechanisms.

Study Design

The authors examined five patients with extensive corneal neovascularization arising from:

- Infectious keratitis
- Limbal stem cell deficiency
- Interstitial keratitis

Anterior-segment OCT scans were obtained using a Heidelberg Spectralis OCT system.

For each patient:

- Twenty-one scans were acquired across the superior cornea
- Twenty-one scans were acquired across the inferior cornea
- Forty-two scans were analyzed per eye

OCT findings were compared against slit-lamp examination and clinical imaging.

Description of the Barcode Sign

Major vascular trunks within the cornea completely obstructed OCT light transmission.

This produced:

- A vertical hypo-reflective stripe
- Extension of the stripe posterior to the vessel
- Stripe width proportional to vessel width
- Multiple parallel stripes when several vessels were present

When many vessels were visualized simultaneously, the OCT image resembled a retail barcode pattern, leading to the term "Barcode Sign" where the stripes appeared as dark linear regions embedded within the brighter OCT background.

Proposed Mechanism

The authors hypothesized that the shadowing effect arises primarily from the blood column within actively perfused vessels rather than the vessel wall itself.

Evidence supporting this interpretation includes:

- Both afferent and efferent vessels generated equivalent shadows
- Shadow intensity did not appear dependent on vessel-wall structure
- Ghost vessels lacking active blood flow did not generate shadows

Therefore, the attenuation pattern appears linked to the presence of circulating blood. The barcode sign may thus represent an OCT manifestation of active vascular perfusion.

Relationship to OCT Physics

The barcode sign illustrates a general OCT principle: Structures that strongly attenuate or absorb OCT light can generate vertical shadow artifacts extending into deeper tissue layers.

The resulting pattern consists of:

- Localized signal loss
- Vertical transmission defects
- Stripe-like image structures
- Reduced visibility of deeper anatomy

Such shadowing artifacts are well known in OCT imaging and can reveal information about the underlying structure causing attenuation.

Relevance to AMD Progression Research

Although the barcode sign was identified in corneal vascularization, the study is conceptually relevant to AMD imaging. Several advanced AMD features can alter OCT signal transmission, including:

- Retinal pigment epithelium (RPE) degeneration
- Geographic atrophy
- Hypertransmission defects
- Choroidal structural abnormalities
- Neovascular complexes

These abnormalities can create vertically organized intensity patterns within OCT volumes. Consequently, the barcode sign provides evidence that biologically meaningful stripe-like structures may emerge naturally from tissue-specific optical interactions rather than image-processing artifacts.

Implications for the Current Project

The AMD progression project investigates whether stripe-like "barcoding" patterns visible within OCT scans are associated with future disease progression.

The barcode sign literature suggests several hypotheses:

- Stripe patterns may arise from localized attenuation or transmission phenomena.
- Stripe width may reflect properties of the underlying anatomical structure.
- Stripe density may provide information about disease burden.
- Quantitative characterization of stripe morphology may produce useful imaging biomarkers.

Potential quantitative descriptors include stripe count, spacing, width, contrast, orientation, or spatial frequency content.

While the biological mechanism in AMD may differ from corneal vascular shadowing, this study provides an important precedent demonstrating that barcode-like OCT patterns can arise from meaningful underlying tissue structure and physiology.

3. Self-Attention CNN for Retinal Layer Segmentation

Overview

Cao et al. proposed a hybrid convolutional neural network (CNN) and Transformer architecture for retinal layer segmentation in optical coherence tomography (OCT) images. The primary objective was to improve segmentation accuracy by combining the local feature extraction capabilities of CNNs with the long-range dependency modeling provided by self-attention mechanisms.

The proposed architecture extends the U-Net framework by incorporating:

- A Transformer-based self-attention module,
- Attention Gates (AGs),
- Channel attention mechanisms,
- A one-dimensional attention strategy tailored to retinal OCT anatomy.

The model was evaluated on public retinal OCT datasets and demonstrated improved segmentation performance compared to several existing CNN-based and Transformer-based approaches.

Motivation

Retinal OCT images possess highly structured anatomical organization. Traditional CNNs perform well at extracting local image features but have limited receptive fields, making it difficult to capture long-range spatial relationships.

Challenges in OCT segmentation include layer boundaris spanning large portions of the image, anatomical variability, pathology-induced distortion, and loss of contextual information during downsampling.

The authors argue that successful segmentation requires both local texture information and global structural context. Self-attention mechanisms are introduced to address this limitation.

Self-Attention Mechanism

The Transformer component allows image regions to interact with one another regardless of spatial distance. Self-attention is computed as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V,$$

where Q = query matrix, K = key matrix, V = value matrix, and d = feature dimension.

Unlike convolution, which uses local neighborhoods, self-attention provides a global receptive field. This allows the model to learn relationships between distant retinal structures and improve segmentation consistency across the image.

One-Dimensional Transformer Design

A key innovation of the paper is the use of a one-dimensional Transformer. The authors observed that retinal anatomy is organized primarily along the vertical direction. Instead of computing attention across the entire two-dimensional image, feature maps are rearranged so that attention operates only along the retinal depth dimension.

Feature maps are reshaped from

$$(B, C, H, W) \rightarrow (BW, H, C)$$

where B = batch size, C = number of channels, H = image height, and W = image width.

This design preserves anatomical structure, reduces computational cost, increases effective receptive field, and improves efficiency relative to full 2D attention.

Encoder–Decoder Architecture

The overall architecture follows the encoder–decoder paradigm of U-Net.

Encoder

The encoder consists of repeated blocks of:

- 3×3 convolutions,
- Batch normalization,
- ReLU activation,
- Max pooling.

As depth increases, spatial resolution decreases and feature complexity increases. The encoder extracts increasingly abstract representations of retinal structures.

Decoder

The decoder performs:

- Upsampling,
- Feature fusion,
- Boundary reconstruction,
- Pixel-wise segmentation.

Skip connections transfer high-resolution information from encoder layers to corresponding decoder layers. These connections help preserve fine retinal boundary details that may otherwise be lost during downsampling.

Attention Gates

The model incorporates Attention Gates (AGs) within skip connections. Rather than passing all encoder features directly to the decoder, attention mechanisms determine which features are most relevant.

Conceptually:

$$\tilde{x} = \alpha x, \quad \forall \quad 0 \leq \alpha \leq 1$$

where α is a learned attention coefficient.

Benefits include regularization (suppression) of irrelevant features, improved focus on retinal boundaries, and enhanced segmentation precision.

Channel Attention

In addition to spatial attention, channel attention is used to weight feature channels according to importance. Different channels often capture different types of information:

- Edges,
- Layer boundaries,
- Texture patterns,
- Anatomical structures.

Channel attention allows the network to emphasize informative channels and reduce the influence of less useful features.

Loss Function

Training uses a weighted combination of Dice loss and multiclass cross-entropy loss.

The Dice coefficient is

$$\text{Dice} = \frac{2|X \cap Y|}{|X| + |Y|}.$$

Corresponding Dice loss L_{Dice} and cross-entropy loss L_{CE} are:

$$L_{\text{Dice}} = 1 - \text{Dice}. \quad L_{\text{CE}} = -\sum_i y_i \log(p_i).$$

The overall objective is

$$L = \lambda_1 L_{\text{Dice}} + \lambda_2 L_{\text{CE}}.$$

Combining the two losses balances pixel level calssification accuracy and global segmentation overlap.

Datasets

The model was evaluated on two retinal OCT datasets:

- DUKE Diabetic Macular Edema (DME) Dataset,
- Optic Disc Retinal OCT Dataset.

These datasets contain manually annotated retinal layer segmentations used as ground truth labels.

Results

Performance was evaluated using the Dice coefficient. For the DUKE DME dataset, an average score of 0.871 was reported, and for the optic-disc dataset, 0.820 was reported. This outperforms UNet, ATtention UNet, TransUNet, RelayNet, and DRUNey. The improvements suggest that combining CNN-based local feature extraction with Transformer-based global context modeling can improve retinal layer segmentation performance.

Key Takeaways

The primary contribution of this work is the integration of self-attention into a U-Net style segmentation framework for retinal OCT images.

Important observations include:

- CNNs effectively learn local retinal texture and boundary information.
- Self-attention provides global contextual information and long-range dependency modeling.
- Restricting attention to the vertical dimension leverages the layered anatomical structure of retinal OCT images.
- Attention Gates improve feature selection during decoding.
- Combining local and global representations improves segmentation accuracy relative to standard CNN architectures.

The paper demonstrates the potential value of hybrid CNN–Transformer architectures for retinal OCT analysis and segmentation tasks.

4. Machine Learning for Detection of Geographic Atrophy in Spectral OCT

This paper presents two deep learning pipelines for automated detection and quantification of geographic atrophy (GA) from spectral-domain optical coherence tomography (SD-OCT). Rather than relying on fundus autofluorescence (FAF), the authors use volumetric OCT data to identify regions of retinal pigment epithelium (RPE) loss and associated hypertransmission defects. Although the work focuses on supervised semantic segmentation, several of the ideas are directly applicable to our goal of automatically identifying and quantifying barcoding artifacts.

4.1. Clinical Motivation

Geographic atrophy represents the advanced stage of dry age-related macular degeneration (AMD), characterized by degeneration of the retinal pigment epithelium (RPE), photoreceptors, and outer retinal layers. Progression to GA is irreversible and results in permanent vision loss.

The Classification of Atrophy Meetings (CAM) consensus defines complete RPE and outer retinal atrophy (cRORA) on OCT by the simultaneous presence of

- complete RPE loss,
- overlying photoreceptor degeneration, and
- hypertransmission into the choroid,

with each lesion spanning at least $250\,\mu$m in its greatest diameter.

The authors argue that OCT is preferable to fundus autofluorescence because it provides three-dimensional structural information and allows direct visualization of retinal layers rather than only surface appearance.

4.2. Ground Truth Construction

A particularly interesting aspect of the work is that the authors do not manually delineate every GA lesion from scratch. Instead, they leverage retinal layer segmentation.

Previously validated machine learning models segment three retinal boundaries:

- Ellipsoid Zone (EZ)
- Retinal Pigment Epithelium (RPE)
- Bruch's Membrane (BM)

Ground-truth GA masks are then generated automatically wherever these three segmented layers completely overlap, indicating total disappearance of both the EZ and RPE.

Conceptually,

$$\text{GA} \iff EZ = RPE = BM$$

within a local region.

This produces pixel-level training masks while dramatically reducing manual annotation effort.

4.3. B-Scan versus En Face Models

The paper develops two complementary segmentation models.

The first operates directly on individual OCT B-scans. This approach detects GA using cross-sectional structural information, making it highly specific to retinal anatomy. Because every slice is analyzed independently, subtle structural changes are preserved.

The second operates on en face OCT projections. Instead of detecting retinal layer loss directly, the model identifies regions of hypertransmission visible in the en face image. Predictions can then be projected back onto individual B-scans using the Bruch's membrane segmentation.

The authors emphasize that the two representations provide complementary information:

- B-scans capture detailed retinal anatomy.
- En face images capture global lesion morphology.

Future work proposed by the authors is to combine both models into a hybrid system.

4.4. Model Architecture

Both models employ a standard U-Net architecture for semantic segmentation.

Training used

- resized 256×256 inputs,
- binary cross-entropy loss,
- approximately twenty million trainable parameters,
- automatic hard-example retraining by oversampling poorly performing patches.

Although technically straightforward, the authors demonstrate that careful construction of the training labels contributes more to performance than architectural novelty.

4.5. Results

The B-scan model achieved

- 91% detection accuracy,
- 94% pixel-wise segmentation accuracy,
- ICC = 0.88 for lesion measurements.

The en face model achieved

- 82% detection accuracy,
- 96% pixel-wise segmentation accuracy,
- ICC = 0.95 for lesion measurements.

Performance generalized well across both Heidelberg Spectralis and Zeiss Cirrus OCT systems, suggesting that the learned structural features are relatively device independent.

4.6. Hypertransmission as an Imaging Biomarker

Perhaps the most interesting observation is that many apparent false positives from the en face model were not actually incorrect predictions.

Instead, the model frequently identified isolated hypertransmission defects that did not yet satisfy the complete CAM definition of geographic atrophy.

Consequently, the authors hypothesize that hypertransmission itself may represent a useful biomarker preceding fully developed GA. Rather than suppressing these predictions, they suggest explicitly modeling hypertransmission independently from established GA.

This observation is particularly relevant because our project is similarly interested in identifying image features that may precede disease progression rather than only detecting already established pathology.

4.7. Relevance to Barcoding Detection

Although the objective differs substantially from ours, several ideas transfer directly.

First, the paper demonstrates that OCT contains sufficient structural information for automated discovery of clinically meaningful biomarkers.

Second, the authors emphasize representation learning directly from image structure rather than relying solely on handcrafted measurements.

Third, the work illustrates how clinically meaningful imaging biomarkers can be quantified automatically once reliable image representations have been obtained.

Our problem differs in one important respect. Rather than detecting a well-defined pathological structure such as GA, we seek to characterize vertical barcoding artifacts whose morphology is not currently described by a precise anatomical definition. Consequently, constructing exhaustive pixel-level annotations suitable for supervised segmentation would be difficult.

Instead, our objective is to develop an unsupervised representation capable of identifying and quantifying recurring vertical stripe patterns without requiring dense manual labels.

4.8. Limitations

From the perspective of the present project, several limitations remain.

- The approach requires extensive pixel-level ground truth.
- Performance depends upon accurate retinal layer segmentation.
- U-Net learns supervised anatomical labels rather than discovering novel biomarkers.
- The framework detects predefined pathology instead of identifying previously unknown structural patterns.

These limitations motivate investigating unsupervised feature learning approaches for quantitative characterization of barcoding.

4.9. Method Sketch

Inspired by the OCT-based workflow of Kalra et al., our approach similarly begins by leveraging the anatomical structure available within individual B-scans rather than attempting to analyze the entire OCT image directly. However, instead of performing supervised semantic segmentation of geographic atrophy, our objective is to discover and quantify barcoding patterns using unsupervised image analysis.

The proposed pipeline consists of four stages.

First, each Heidelberg Spectralis B-scan is paired with the retinal layer segmentations provided by the HEYEX software. These segmentations are used to identify the Bruch's membrane (BM), which serves as a stable anatomical reference. Each B-scan is flattened with respect to the BM in order to remove normal retinal curvature and standardize image geometry across subjects.

Following flattening, only the image region at and below the BM is retained. This isolates the portion of the OCT image where hypertransmission and barcoding are expected to occur while removing the majority of unrelated retinal structure. Restricting the analysis to this anatomically meaningful region substantially reduces background variation and focuses subsequent processing on tissue most relevant to the proposed biomarker.

Within this region of interest, an unsupervised algorithm will be used to identify candidate barcoding regions based on image appearance. Rather than relying upon manually defined annotations, the method will learn or estimate characteristic intensity patterns associated with barcoding. Candidate features include increased local brightness, oscillatory vertical intensity profiles, periodic texture, or combinations thereof. Multiple unsupervised approaches will be explored, and the final representation will be selected based on robustness and consistency across patients.

Instead of producing a dense pixel-level segmentation mask, the initial objective is to identify the horizontal extent of barcoding along each B-scan. The algorithm should therefore produce an overlay indicating regions classified as normal and regions exhibiting barcoding, for example,

$$\underbrace{\qquad\qquad\qquad}_{\text{Normal}}\;\underbrace{\qquad\qquad\qquad}_{\text{Barcoding}}\;\underbrace{\qquad\qquad}_{\text{Normal}}$$

where the highlighted interval corresponds to the automatically detected barcoding region.

Once the affected region has been localized, quantitative biomarkers can be extracted exclusively within the detected interval. Candidate measurements include the horizontal extent of barcoding, average and maximum brightness, vertical intensity oscillation statistics, texture descriptors, and additional quantitative measures that may correlate with disease progression. By separating localization from quantification, the framework naturally supports future refinement of both detection algorithms and downstream biomarkers without altering the overall pipeline.

Part II

Project Directions

5. Barcoding Detection and Quantification

The aim now is to figure out whether V_2 (Baseline) predicts Early Atrophy Progression or not. To do so, we will take the five fastest progressors and five slowest progressors and create a ground truth by marking EA and barcoding regions.

5.1. Data Preparation

We first take the five fastest progressors and five slowest progressors and create a ground truth by marking EA and barcoding regions. We will then use this ground truth to train a model to detect barcoding regions in new images.

1. Double click patient
2. Go to V2 HRA OCT and select the smallest scan (20x20)
3. Got to Display → Overlay → Measure Distance
4. Lines must be flat, mark the barcoding regions (RED) and EA regions (YELLOW)
5. Image → Export as picture (JPG) → Add Overlays

5.2. Future Work

Following creation of the ground truth dataset, several directions will be investigated.

First, the manually annotated images will be analyzed to understand the spatial relationship between Early Atrophy (EA) and barcoding regions, as well as differences between fast and slow progressors. EDA will guide the choice of modeling approach and help determine whether the annotations are sufficiently consistent for automation.

ML methods capable of localizing clinically relevant regions will be explored such as deep learning models for localization and semantic segmentation, weakly supervised methods that leverage the clinician annotations, and existing OCT analysis methods from the literature. The objective is to produce an interpretable overlay that highlights predicted EA and barcoding regions on previously unseen baseline OCT scans.

Once reliable localization has been achieved, quantitative measurements of barcoding will be investigated. Potential biomarkers include the horizontal extent of barcoding, the number of distinct barcode regions, total barcode coverage, overlap with Early Atrophy, and the spatial relationship between EA and barcoding.

Finally, the developed methods will be validated against the clinician-generated ground truth to assess localization accuracy, measurement consistency, and clinical interpretability.

6. Extraction and Measurement of Manual Markings

To establish a quantitative baseline for subsequent image analysis, a manual annotation dataset was first constructed from optical coherence tomography (OCT) scans. Representative cases corresponding to the fastest and slowest progressing eyes were identified by the clinical collaborators. Using the Heidelberg HEYEX software, retinal specialists manually annotated two clinically relevant features on selected B-scans:

1. Regions exhibiting barcoding artifacts, and

2. Regions corresponding to external atrophy (EA).

These annotations were exported as JPEG images containing the original OCT scan overlaid with colored measurement segments. Red and yellow measurement lines were used to distinguish between the two annotation types, while the numerical measurement values remained embedded only within the HEYEX interface. Consequently, an automated procedure was required to recover the annotation geometry directly from the exported images.

6.1. Automatic Extraction of Manual Annotations

The objective of the extraction algorithm was to identify every manually drawn measurement segment and estimate its length in image pixels. Because the exported JPEGs contain interface elements, text labels, and other colored graphics in addition to the OCT image itself, the extraction procedure consisted of several stages designed to progressively isolate the clinician-generated markings.

The first step was localization of the OCT B-scan within the exported image. Rather than processing the full screenshot, the OCT display panel was cropped using fixed interface boundaries, substantially reducing the search space and eliminating unrelated graphical elements.

Within the cropped OCT image, colored annotations were isolated using HSV thresholding. Separate color masks were constructed for the red and yellow overlays, producing binary images containing only candidate annotation pixels. Since the measurement segments are predominantly horizontal, morphological filtering with a horizontal structuring element was applied to suppress isolated noise while reinforcing elongated horizontal structures.

Connected-component analysis was subsequently performed on each binary mask. Initially, individual measurement lines frequently appeared fragmented into multiple connected components. Investigation revealed that this fragmentation was not caused by breaks in the annotations themselves, but rather by nearby measurement text that was rendered in the same color as the corresponding annotation. The text labels were therefore detected as additional connected components.

To address this issue, a geometric filtering procedure was developed. Components satisfying the following criteria were removed:

- located vertically above another component,

- sufficiently close in vertical distance,

- overlapping horizontally with a longer component, and

- substantially shorter than the underlying horizontal segment.

This heuristic successfully removed measurement text while preserving the underlying annotation. Remaining adjacent horizontal components were then merged into single measurement segments, yielding one connected component per clinician annotation.

For each recovered segment, the algorithm recorded the horizontal start coordinate, end coordinate, color classification, and pixel length. These measurements formed the basis of the subsequent quantitative analysis.

6.2. Toward Physical Measurement Recovery

Although pixel lengths can be recovered directly from the exported images, the clinically relevant quantities are expressed in microns within the HEYEX software. To recover physical distances, metadata stored within the proprietary Heidelberg `.E2E` files was investigated using the `eyepy` package.

Each B-scan provides acquisition metadata including image dimensions, scan positions, acquisition time, and several calibration-related fields. In particular, the metadata contains multiple candidate quantities describing scan geometry, including scan endpoints, image dimensions, and undocumented calibration parameters.

An initial investigation suggested that one undocumented metadata field may correspond to a physical scan width. However, this quantity varied across B-scans despite nearly identical acquisition geometry, indicating that its interpretation remains unclear. Consequently, a reliable conversion from exported JPEG pixel lengths to physical distances could not yet be established using metadata alone.

At present, pixel-based measurements can be recovered automatically, while the mapping to physical units remains an open calibration problem requiring further investigation.

6.3. Future Investigation

Several avenues remain for improving the annotation extraction pipeline. First, additional validation will be performed across a substantially larger collection of manually annotated scans to assess robustness under varying image quality, annotation density, and acquisition settings.

Second, alternative strategies for recovering physical measurements will be explored. Rather than relying solely on undocumented metadata fields, future work will investigate direct calibration using native OCT image geometry, known scan dimensions, or empirical calibration against measurements reported within the HEYEX software.

Finally, the extracted annotation dataset will serve as ground truth for subsequent automated segmentation models, providing paired image–measurement data for learning clinically relevant biomarkers directly from OCT scans.